

25. Consider the following decimal numbers:

- I. 1.1666666... II. 1.181181118...
 III. 2.010010001... IV. 1.454545...

Which of the above numbers represent(s) rational number(s)?

- (a) Only IV (b) II and III
 (c) I and IV (d) None of these

PREVIOUS YEARS' QUESTIONS

26. What is the value of $0.00\overline{7} + 17.\overline{83} + 310.020\overline{2}$?

- (a) $327.86\overline{638}$ (b) $327.86\overline{638}$ **✓ 2012 I**
 (c) $327.86\overline{683}$ (d) $327.86\overline{68}$

27. What is the value of $0.242424...$? **✓ 2012 II**

- (a) $\frac{23}{99}$ (b) $\frac{8}{33}$ (c) $\frac{7}{33}$ (d) $\frac{47}{198}$

28. Representation of $0.\overline{2341}$ in the form $\frac{p}{q}$, where p and q are integers, $q \neq 0$, is **✓ 2013 I**

- (a) $\frac{781}{3330}$ (b) $\frac{1171}{4995}$ (c) $\frac{2341}{9990}$ (d) $\frac{2339}{9990}$

29. Let p be a prime number other than 2 or 5. One would like to express the vulgar fraction $1/p$ in the form of a recurring decimal. Then, the decimal will be **✓ 2015 I**

- (a) a pure recurring decimal and its period will be necessarily $(p-1)$
 (b) a mixed recurring decimal and its period will be necessarily $(p-1)$
 (c) a pure recurring decimal and its period will be some factor of $(p-1)$
 (d) a mixed recurring decimal and its period will be some factor of $(p-1)$

30. The value of $(0.\overline{63} + 0.\overline{37})$ is **✓ 2015 II**

- (a) 1 (b) $\frac{100}{91}$ (c) $\frac{100}{99}$ (d) $\frac{1000}{999}$

ANSWERS

1		2		3		4		5		6		7		8		9		10	
11		12		13		14		15		16		17		18		19		20	
21		22		23		24		25		26		27		28		29		30	

HINTS AND SOLUTIONS

1. (a) $x = 15.9273 - 11.0049 = 4.9224$

2. (d)
$$\begin{array}{r} 15.3900 \\ 0.2360 \\ 5.2900 \\ + 0.0002 \\ \hline 20.9162 \end{array}$$

 $x = 20.9162$

3. (a) $1.75 \times 1.24 = 2.17$
 $\Rightarrow 1.75 \times 124 = 217$

4. (a) Since, $\frac{111.744}{28.8} = 3.88$
 $\Rightarrow \frac{111744}{28800} = 3.88$ [Rule 6]

Now, $\frac{1117.44}{288} = \frac{11744}{28800} = 3.88$ [Rule 4]

5. (b)
$$\frac{[(0.5)^2]^2 - [(0.4)^2]^2}{(0.5)^2 + (0.4)^2}$$

$$[(a^2 - b^2) = (a + b)(a - b)]$$

$$\begin{aligned} &= \frac{[(0.5)^2 - (0.4)^2][(0.5)^2 + (0.4)^2]}{[(0.5)^2 + (0.4)^2]} \\ &= (0.5)^2 - (0.4)^2 = (0.5 - 0.4)(0.5 + 0.4) \quad [\text{Rule 1}] \\ &= 0.1 \times 0.9 = 0.09 \quad [\text{Rule 2}] \end{aligned}$$

6. (b)
$$\frac{0.004 \times 0.0008}{0.002} = \frac{0.000032}{0.02}$$

 $= 0.00016$

7. (a) Here, $\frac{6}{8} = 0.75$, $\frac{7}{9} = 0.\overline{7}$, $\frac{5}{6}$
 $= 0.8\overline{3}$, $\frac{11}{13} = 0.846$
 So, $0.75 < 0.\overline{7} < 0.8\overline{3} < 0.846$

8. (c) Given, $0.232323... = 0.\overline{23}$
 (which is a recurring decimal) $= \frac{23}{99}$

9. (b) Here, $10^k = \frac{0.0003245}{3.245}$

$$= \frac{3.245 \times 10^{-4}}{3.245}$$

$10^k = 10^{-4}$
 So, $k = -4$

10. (c) $0.\overline{6} + 0.\overline{8} + 0.\overline{7} = \frac{6}{9} + \frac{8}{9} + \frac{7}{9}$
 $= \frac{21}{9} = \frac{7}{3} = 2\frac{1}{3}$

11. (a) Here, $6.\overline{88} - 2.\overline{58}$
 $= \left(6 + \frac{88}{99}\right) - \left(2 + \frac{58}{99}\right)$
 $= (6 - 2) + \left(\frac{88}{99} - \frac{58}{99}\right)$
 $= 4 + \frac{88 - 58}{99} = 4 + \frac{30}{99} = 4.\overline{30}$

12. (c) Given expression
 $= (1198)^2 + (0.02)^2 + 1198 \times x$
 For the given expression to be a perfect square, we must have
 $1198 \times x = 2 \times 1198 \times 0.02$
 $\Rightarrow x = 0.04$
 [by using $(a + b)^2 = a^2 + b^2 + 2ab$]

$$13. (b) 2.5x = 0.5y \\ \Rightarrow \frac{x}{y} = \frac{0.5}{2.5} = 0.20$$

Now, the expression is [dividing numerator and denominator by y]

$$\frac{x+y}{x-y} = \frac{x/y+1}{x/y-1} = \frac{0.20+1}{0.20-1} = \frac{1.20}{-0.80} = -15$$

Shortcut Method

$$\frac{x}{y} = \frac{0.5}{2.5} \Rightarrow \frac{x+y}{x-y} = \frac{0.5+2.5}{0.5-2.5}$$

[using componendo and dividendo]

$$\Rightarrow \frac{x+y}{x-y} = \frac{3}{-2} = -1.5$$

$$14. (c) 7.2 - \frac{7.2}{10} \\ \Rightarrow 7.2 - 0.72 = 6.48$$

$$15. (d) \text{ Given, } \frac{p}{q} = 2.5\overline{2} \quad \dots(i)$$

Now, 100 multiply both sides, we get

$$\frac{100p}{q} = 252.\overline{2} \quad \dots(ii)$$

On subtracting Eq. (i) from Eq. (ii), we get

$$\frac{99p}{q} = 250 \Rightarrow \frac{p}{q} = \frac{99}{250} = 0.396$$

$$16. (c) \text{ Required decimal fraction} = \frac{1}{60 \times 60} \\ = \frac{1}{3600} = 0.00027$$

$$17. (a) \frac{3\sqrt{5}}{2\sqrt{5} - 0.48} = \frac{3 \times 2.24}{2 \times 2.24 - 0.48} \\ [\because \sqrt{5} = 2.24] \\ = \frac{6.72}{4.48 - 0.48} = \frac{6.72}{4} = 1.68$$

$$18. (d) \text{ Here, } \left[\frac{(0.1)^2 - (0.01)^2}{0.0001} + 1 \right] \\ = \frac{0.01 - 0.0001}{0.0001} + 1 = \left(\frac{0.0099}{0.0001} + 1 \right) \\ = (99 + 1) = 100$$

$$19. (b) \frac{3}{3 + 0.3 - 3.03} = \frac{3}{3 - 2.73} \\ = \frac{3}{3 \times 0.91} = \frac{3}{3 \times 91}$$

$$= \frac{3}{3-1} = \frac{3}{2} = 1.5$$

$$20. (b) \text{ Let } x \text{ should be subtracted} \\ (0.527 \times 2.013) - x > 1 \\ 1.060851 - x > 1 \\ x = 0.060851$$

$$21. (b) 0.34\overline{67} + 0.13\overline{33} \\ = \frac{3467 - 34}{9900} + \frac{1333 - 13}{9900} \\ = \frac{3433 + 1320}{9900} = \frac{4753}{9900} \\ = \frac{4801 - 48}{9900} = 0.48\overline{01}$$

$$22. (c) \frac{2}{5} = 0.4, \frac{5}{6} = 0.8\overline{3}, \frac{11}{12} = 0.91\overline{6} \\ \text{and } \frac{7}{8} = 0.875$$

Clearly, the greatest fraction is $0.91\overline{6}$ i.e. $\frac{11}{12}$.

$$23. (c) \frac{3}{4} = 0.75, \frac{5}{6} = 0.83\overline{3} \\ \frac{1}{2} = 0.5, \frac{2}{3} = 0.6\overline{6}, \frac{4}{5} = 0.8 \text{ and } \frac{9}{10} = 0.9 \\ \text{Clearly, } 0.8 \text{ lies between } 0.75 \text{ and } 0.83\overline{3}. \\ \therefore \frac{4}{5} \text{ lies between } \frac{3}{4} \text{ and } \frac{5}{6}.$$

$$24. (c) \text{ In } \frac{1}{22} \text{ and } \frac{2}{15}, 22 \text{ and } 15 \text{ are not in} \\ \text{the form of } 2^m \times 5^n \text{ but in } \frac{1}{16}. \\ 16 \text{ in the form of } 2^4 \times 5^0. \text{ So, } \frac{1}{16} \text{ can be} \\ \text{written as a terminating decimal.} \\ \text{Hence, statements I and III are correct.}$$

25. (c) Since, $1.16666\dots$ and $1.454545\dots$ are recurring numbers and we know that, recurring numbers represent rational numbers.

Hence, statements I and IV are rational numbers.

$$26. (b) 0.00\overline{7} = 0.007777777 \\ \Rightarrow 17.\overline{83} = 17.83838383 \\ 310.020\overline{2} = 310.020222222 \\ = 327.8663838 \text{ on adding} \\ = 327.866\overline{38}$$

$$27. (b) \text{ Given, } 0.242424\dots = 0.2\overline{4} \\ [\text{which is a recurring decimal number}] \\ = \frac{24}{99} = \frac{8}{33}$$

$$28. (d) 0.23\overline{41} = \frac{2341 - 2}{9990} = \frac{2339}{9990}$$

$$29. (c) \text{ Value } p \text{ may be } 3, 7, 11, 13. \\ \frac{1}{3} = 0.\overline{3}, \text{ Period} = 1$$

Here, $p-1=2$ and 1 is a factor of 2.

$$\frac{1}{7} = 0.1428\overline{57}, \text{ Period} = 6$$

Here, $p-1=6$ and 6 is a factor of 6.

$$\frac{1}{11} = 0.09\overline{09}, \text{ Period} = 2$$

Here, $p-1=10$ and 2 is a factor of 10.

$$30. (c) \text{ We have, } 0.6\overline{3} + 0.3\overline{7}$$

Let $x = 0.636363\dots$

$$100x = 63.636363\dots, 99x = 63$$

$$\text{and } x = \frac{63}{99}$$

$$\text{Similarly, } y = 0.3\overline{7} = 0.373737$$

$$\therefore y = \frac{37}{99}$$

$$\therefore x + y = \frac{63}{99} + \frac{37}{99} = \frac{100}{99}$$

05

SQUARE ROOTS AND CUBE ROOTS

Generally (1-3) questions have been asked from this chapter. This section will be useful in solving simplification questions and will save lots of time while doing fuzzy calculations.



SQUARE

The square of a number is obtained by multiplying the number by itself.

e.g. Square of $5 = 5 \times 5 = 25$, square of $6 = 6 \times 6 = 36$

Perfect square A natural number n is called a perfect square or a square number, if there exists a natural number m such that $n = m^2$.

e.g. The numbers 1, 4, 9, 16, 25, ... are perfect square.

Properties of Perfect Square

The properties of perfect squares are discussed below.

- (i) A number having 2, 3, 7 or 8 at unit's place is never a perfect square.
e.g. 172, 2783
- (ii) The squares of odd numbers are odd and the squares of even numbers are even.
- (iii) The difference of squares of two consecutive natural numbers is equal to their sum.
e.g. $7^2 - 6^2 = 7 + 6 = 13$
- (iv) The product of four consecutive natural numbers plus one is always a square.
- (v) A number ending in an odd number of zeroes is never a perfect square.

Square Root

The square root of a number x is that number which when multiplied by itself gives x as the product. It is denoted by $\sqrt{\quad}$.

In general, $y^2 = x \Rightarrow y = \sqrt{x}$

Here, y is the square root of x , if and only if x is the square of y .

e.g. $\sqrt{4} = \sqrt{2 \times 2} = 2$

Thus, 2 is the square root of 4.

EXAMPLE 1. If $\sqrt{\frac{x}{64}} = \frac{4}{8}$, then the value of x is

- a. 16 b. 12 c. 8 d. 4

Sol. a. We have, $\sqrt{\frac{x}{64}} = \frac{4}{8}$,

On squaring both sides, we get

$$\left(\sqrt{\frac{x}{64}}\right)^2 = \left(\frac{4}{8}\right)^2 \Rightarrow \frac{x}{64} = \frac{16}{64} \Rightarrow x = 16$$

Methods to Find Square Root

The square root of a given number can be determined by using any of the following two methods

1. Prime Factorization Method
2. General Method/Division Method

Prime Factorization

In this method, we express the given number as the product of prime factors. Now, for finding square root, we take the product of these prime factors choosing one out of every pair of same prime factors.

EXAMPLE 2. The square root of 213444 is

- a. 332 b. 368 c. 432 d. 462

Sol. d. The prime factorisation of 213444 is

$$213444 = 2 \times 2 \times 3 \times 3 \times 7 \times 7 \times 11 \times 11$$

$$\therefore \sqrt{213444} = \sqrt{2 \times 2 \times 3 \times 3 \times 7 \times 7 \times 11 \times 11}$$

Now, taking one number from each pair and multiplying them, we get $\sqrt{213444} = 2 \times 3 \times 7 \times 11 = 462$

EXAMPLE 3. What is the square root of $\frac{0.324 \times 0.64 \times 129.6}{0.729 \times 1.024 \times 36}$?

- a. 4 b. 3
c. 2 d. 1

Sol. d.
$$\sqrt{\frac{0.324 \times 0.64 \times 129.6}{0.729 \times 1.024 \times 36}} = \sqrt{\frac{324 \times 64 \times 1296}{729 \times 1024 \times 36}}$$

$$= \sqrt{\frac{(2^2 \times 3^4) \times (2^6) \times (2^4 \times 3^4)}{(3^6) \times (2^{10}) \times (2^2 \times 3^2)}} = \frac{(2 \times 3^2) \times 2^3 \times (2^2 \times 3^2)}{3^3 \times 2^5 \times (2 \times 3)}$$

$$= \frac{18 \times 8 \times 36}{27 \times 32 \times 6} = 1$$

Division Method

To find the square root of a given number using this method, the following steps are to be followed.

Step I In the given number, place bars over every pair of digits starting with the unit's digit. Each pair and the remaining one digit (if any) on the extreme left is called a period.

Step II Think of the largest number whose square is less than or equal to the first period.

Step III Put the quotient above the period and write the product of divisor and quotient just below the first period.

Step IV Subtract the product of divisor and quotient from the first period and bring down the next period to the right of the remainder. This becomes the new dividend.

Step V Double the quotient and enter it with a blank on its right.

Step VI Guess a largest possible digit to fill the blank which will also become the next digit in the quotient, such that when the new divisor is multiplied to the new quotient the product is less than or equal to the dividend.

Step VII Repeat the above steps till all the periods have been taken up.

Now, the quotient so obtained is the required square root of the given number.

EXAMPLE 4. The square root of 1522756 is

- a. 1182 b. 1222
c. 1234 d. 1334

Sol. c

1	1 52 27 56	1234
1		
22	52	
x2	44	
243	0827	
x3	729	
2464	9856	
x4	9856	
	x	

Hence, the square root of 1522756 is 1234.

EXAMPLE 5. The least number of four digits which is a perfect square is 2012 II

- a. 1204 b. 1024
c. 1402 d. 1420

Sol. b. Here, the greatest three-digit number = 999

31	9 99
3	9
x3	9
61	99
	61
	38

Here, the greatest number of three digits which is perfect square = $999 - 38 = 961 = (31)^2$

So, the smallest four-digit number which is perfect square = $(31 + 1)^2 = (32)^2 = 1024$

CUBE

If a number is multiplied three times with itself, then the result of this multiplication is called the cube of that number.
e.g. $8 = 2 \times 2 \times 2 = 2^3$

Properties of Cube

1. Cubes of all even natural numbers are even and all odd natural numbers are odd.
2. Cubes of the numbers ending in digit 0, 1, 4, 5, 6 and 9 are the numbers ending in the same digit.
3. Cubes of negative integers are negative.
4. Cubes of numbers ending in digits 3 and 7 ends in digit 7 and 3, respectively.

CUBE ROOT

The cube root of a given number is the number whose cube is the given number.

In general, we can say that $a^3 = b$, then cube root of 'b' is 'a'. Cube root of a number 'a' is denoted as $\sqrt[3]{a}$.

e.g. $\sqrt[3]{64} = \sqrt{4 \times 4 \times 4} = 4$. Thus, 4 is the cube root of 64.

EXAMPLE 6. If $\sqrt[3]{\frac{x}{27}} = \frac{5}{3}$, then the value of x is

- a. 125 b. 25 c. 27 d. 9

Sol. a. We have $\sqrt[3]{\frac{x}{27}} = \frac{5}{3}$

On cubing both sides, we get

$$\left(\sqrt[3]{\frac{x}{27}}\right)^3 = \left(\frac{5}{3}\right)^3 \Rightarrow \frac{x}{27} = \frac{125}{27}$$

$$\therefore x = 125$$

Method to Find Cube Root

Method to calculate the cube root of a number is as follow

Prime Factorization Method

In this method, we express the given number as the product of prime factors.

Now, for finding cube root, we take the product of these prime factors choosing one out of every three, from the factors.

EXAMPLE 7. Find the cube root of 373248.

- a. 42 b. 52 c. 62 d. 72

Sol. d. Here, $373248 = 8 \times 8 \times 8 \times 9 \times 9 \times 9$

$$373248 = \overbrace{2 \times 2 \times 2} \times \overbrace{2 \times 2 \times 2} \times \overbrace{2 \times 2 \times 2} \times \overbrace{3 \times 3 \times 3} \times \overbrace{3 \times 3 \times 3} \times \overbrace{3 \times 3 \times 3}$$

$$\Rightarrow \sqrt[3]{373248} = 2 \times 2 \times 2 \times 3 \times 3 = 72$$

Hence, the cube root of 373248 is 72.

POWERS OR EXPONENTS

An expression that represents repeated multiplication of the same factor is called a power. It is written as ' x^n ' and is read as 'x' raised to the power n.

Here, x is called the 'base'.

and n is called the 'exponent'.

e.g. $6^5 = 6 \times 6 \times 6 \times 6 \times 6 = 7776$.

Laws of Exponents

Let $\frac{x}{y}$ be any rational number and m, n be any integers.

Then,

$$(i) \left(\frac{x}{y}\right)^m \times \left(\frac{x}{y}\right)^n = \left(\frac{x}{y}\right)^{m+n}$$

$$(ii) \left(\frac{x}{y}\right)^m \div \left(\frac{x}{y}\right)^n = \left(\frac{x}{y}\right)^{m-n} \quad (iii) \left[\left(\frac{x}{y}\right)^m\right]^n = \left(\frac{x}{y}\right)^{mn}$$

$$(iv) \left(\frac{x}{y}\right)^{-n} = \left(\frac{y}{x}\right)^n \quad (v) \left(\frac{x}{y}\right)^0 = 1$$

EXAMPLE 8. If $2^m + 2^{1+m} = 24$, then the value of m is

- a. 1 b. 2 c. 3 d. 5

Sol. c. Given, $2^m + 2^{1+m} = 24 \Rightarrow 2^m + 2^1 \times 2^m = 24$

$$\Rightarrow 2^m(1 + 2) = 24 \Rightarrow 2^m \times 3 = 24 \Rightarrow 2^m = 8 = 2^3$$

Now, on comparing both sides, we get

$$\therefore m = 3$$

Hence, the value of m is 3.

SURDS OR RADICALS

Let x be a rational number and n be a positive integer, such that $\sqrt[n]{x}$ is irrational, then $\sqrt[n]{x}$ is called a radical or surd of order n and here x is called the radicand.

- A surd of order 2 is called a quadratic surd i.e. $\sqrt{2}, \sqrt{3}$.
- A surd of order 3 is called a cubic surd. i.e. $\sqrt[3]{2}, \sqrt[3]{3}$.
- A surd of order 4 is called a biquadratic surd i.e. $\sqrt[4]{5}, \sqrt[4]{7}$.

Laws of Radicals

The laws of exponents which are applicable to the surds are

$$(i) (\sqrt[n]{x})^n = x$$

$$(ii) \sqrt[n]{xy} = \sqrt[n]{x} \cdot \sqrt[n]{y}$$

$$(iii) \sqrt[n]{\frac{x}{y}} = \frac{\sqrt[n]{x}}{\sqrt[n]{y}}$$

$$(iv) (\sqrt[n]{x})^m = (\sqrt[n]{x^m})$$

$$(v) \sqrt[m]{\sqrt[n]{x}} = \sqrt[mn]{x} = \sqrt[n]{\sqrt[m]{x}}$$